

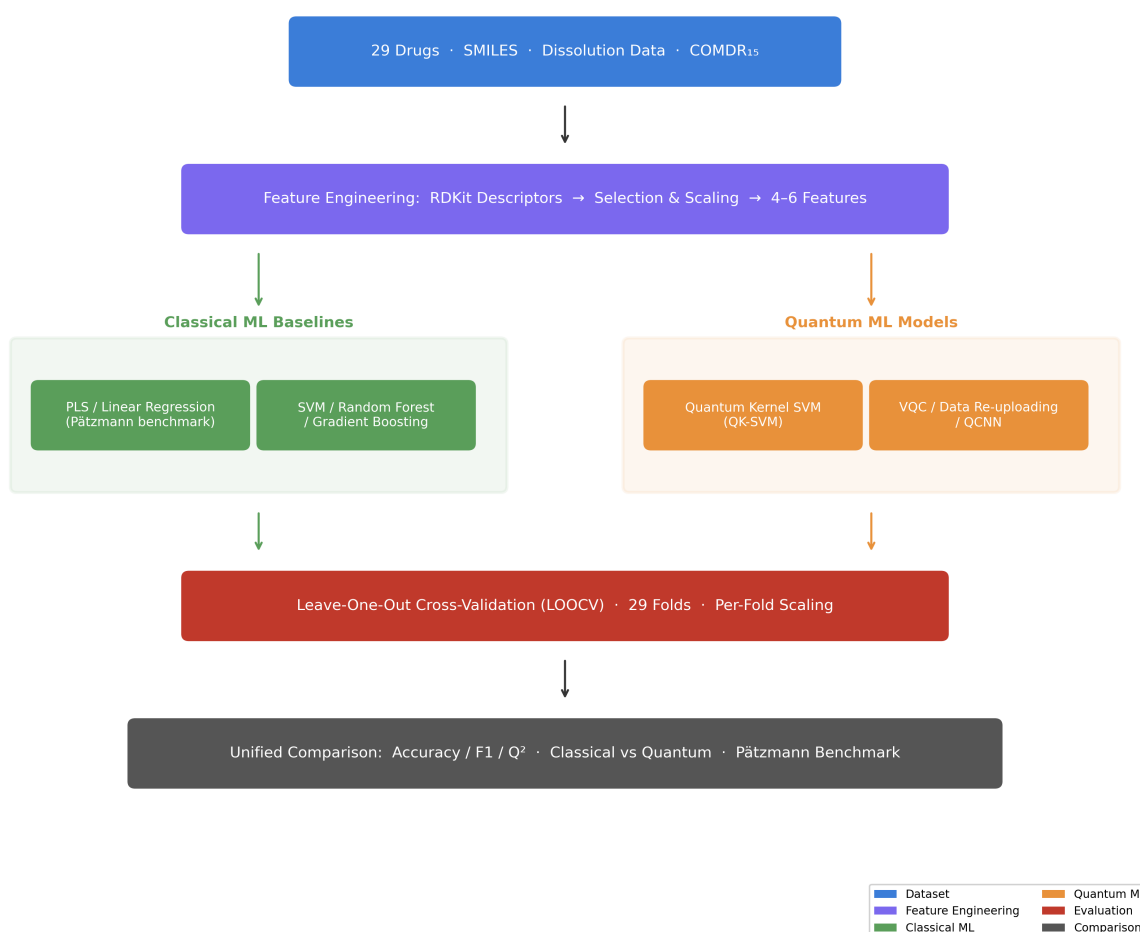
OQI Hackathon 2026

Can (Quantum) Machine Learning Predict Drug Dissolution Enhancement?

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1 Context and Relevance

Nearly **40% of approved drugs and up to 90% of new chemical entities** in development suffer from poor aqueous solubility [2]. When a drug cannot dissolve fast enough in the gastrointestinal tract, it cannot be absorbed, and the patient receives little or no therapeutic benefit regardless of how potent the molecule is. This is one of the biggest bottlenecks in modern drug development.

One industrially established strategy to overcome this hurdle is **co-milling**: the active pharmaceutical ingredient (API) is co-milled together with a polymer excipient, creating an amorphous solid dispersion (ASD) in which the drug is molecularly dispersed, dramatically increasing its apparent solubility and dissolution rate. Not every drug, however, responds equally well to co-milling. Predicting *in advance* which molecules will benefit, and which will not, is currently a slow, expensive, trial-and-error process driven by formulation scientists’ intuition and limited experimental throughput.

This is precisely where **machine learning (ML)** can help. With a growing body of experimental dissolution data and readily computable molecular descriptors, we can ask: *can a model learn from a handful of known drugs to predict the dissolution enhancement of a new one?* And, more ambitiously: *can a quantum ML model do this better than a classical one?*

2 The Dataset and the Reference Classical Model

Experimental Dataset

The dataset used in this challenge was assembled and first published by Pätzmann et al. [1] and covers **29 structurally diverse, crystalline drugs** co-milled with **Polyvinylpyrrolidone K25 (PVP K25)** in a drug-to-excipient ratio of 1:2 (w/w), and characterised in a biorelevant *in vitro* dissolution setup.

The provided data file contains, for each of the 29 drugs:

- D_{50} : median particle size before milling (μm), measured experimentally.
- **Dissolution concentrations** ($\mu\text{g/ml}$) measured at **5, 15, 30, and 60 minutes**, each recorded for four conditions: co-milled formulation (COM), physical mixture (PM), milled drug alone, and unmilled drug alone.

The key response variable is derived from these measurements as the **Co-Milling Dissolution Ratio** at 15 minutes:

$$\text{COMDR}_{15} = \frac{\text{COM}_{15}}{\text{PM}_{15}}, \quad (1)$$

where COM_{15} and PM_{15} are the dissolved drug concentrations at $t = 15$ min for the co-milled formulation and the physical mixture, respectively. A value of $\text{COMDR}_{15} > 1$ indicates that co-milling enhances dissolution; values well above 2 indicate a strong, clinically relevant enhancement. COMDR_{60} is defined analogously using the 60-minute measurements.

Molecular descriptors in the reference paper. Pätzmann et al. computed molecular descriptors using **ADMET Predictor** (Simulations Plus), a commercial software

package, and selected the four most predictive variables via stepwise variable selection: D_{50} (experimental), $\log D_{6.5}$ (lipophilicity at intestinal pH), κ_3 (Kappa shape index), and S_{app} (apparent solubility in FaSSIF).

What participants will receive. For the hackathon, the **SMILES string** of each drug and a **pre-computed table of RDKit descriptors** [9] will be provided. RDKit gives access to more than 200 molecular descriptors, including Lipinski properties, topological indices, shape descriptors, and electronic properties. *Participants must choose which descriptors to use and justify their selection* based on physical reasoning and exploratory analysis. Note that RDKit does not directly compute $\log D_{6.5}$ (which requires a pH-dependent model); $\log P$ (MolLogP) is a readily available approximation. The D_{50} and S_{app} values from the experimental dataset may also be used as features.

The Reference Classical Approach

Pätzmann et al. [1] tackled this problem as a **regression** task: given the physicochemical features of a new drug, predict the continuous value of COMDR_{15} (and COMDR_{60}). Their method of choice was **Partial Least Squares (PLS) regression with variable selection**, a well-established chemometric technique that handles correlated, high-dimensional predictors and small sample sizes gracefully. The best model used exactly **four variables** (D_{50} , $\log D_{6.5}$, κ_3 , S_{app}) selected from a large candidate pool.

Performance was evaluated by LOOCV and reported using R^2 (training fit) and Q^2 (LOOCV predictive power):

Target	R^2 (training)	Q^2 (LOOCV)
COMDR_{15} (PLS, 4 vars)	0.82	0.77
COMDR_{60} (PLS, 4 vars)	0.87	0.84
COMDR_{15} (MLR, 3 vars)	0.83	N/A
COMDR_{60} (MLR, 3 vars)	0.84	N/A

The authors also derived interpretable **multiple linear regression (MLR) equations** using three of the four variables (D_{50} , $\log D_{6.5}$, κ_3), achieving $R^2 = 0.83$ and $R^2 = 0.84$ for the 15 min and 60 min endpoints respectively. The most influential predictor was D_{50} (particle size before milling), followed by $\log D_{6.5}$, κ_3 , and S_{app} .

Key insight from the reference paper

The PLS model achieves $Q^2 = 0.77$ (COMDR_{15}) on only **29 samples** with **4 physicochemical features**: a strong classical chemometric benchmark. The open question is whether a quantum model, exploiting a richer feature space geometry, can *match* this in the regression setting, or *exceed* it in the binary classification reformulation where the decision boundary may be simpler. Note: R^2 (regression) and accuracy (classification) are *not* directly comparable; a $Q^2 = 0.77$ regression model does not imply any particular classification accuracy.

3 Challenge Formulation

Participants are free to approach this problem as either **regression** (predicting the continuous COMDR_{15} value, as in the reference paper) or **binary classification** (predicting whether a drug is a strong responder). For the classification variant, a natural and practically motivated threshold is:

$$y_i = \begin{cases} 1 & \text{(Responder) if } \text{COMDR}_{15}^{(i)} > 2.0 \\ 0 & \text{(Non-Responder) otherwise} \end{cases} \quad (2)$$

A ratio above 2 means co-milling at least doubles the 15-minute dissolution rate, a threshold with direct formulation decision relevance. In the dataset, 13 of the 29 drugs exceed this threshold.

Participants must build **at least two classical ML models** and **at least two quantum ML models**, evaluate all of them under **Leave-One-Out Cross-Validation (LOOCV)**, and present a unified comparison. Results should be interpreted in the context of the Pätzmann benchmark ($Q^2 = 0.77$ for regression; participants establishing their own classification baselines will define the reference bar for the quantum models).

4 Why Quantum Machine Learning?

Classical kernel methods and neural networks are powerful but depend critically on the implicit geometry induced by their kernel or architecture. For a dataset of only 29 samples, classical models face a stark bias–variance dilemma: complex models overfit, simple models underfit.

Quantum ML offers a complementary perspective. By encoding molecular features as quantum states, we implicitly map them into an exponentially large Hilbert space $\mathcal{H}_{2^q}$ where, in principle, decision boundaries that are invisible classically become linearly separable. The encoding map:

$$\mathcal{E} : \mathbb{R}^n \longrightarrow \mathcal{H}_{2^q}, \quad \mathbf{x} \mapsto |\psi(\mathbf{x})\rangle = \bigotimes_{i=1}^q R_y(\pi x_i) |0\rangle, \quad (3)$$

transforms n normalised molecular features into a product state on $q = n$ qubits via **angle encoding**: one R_y rotation per qubit, with feature values scaled to $[0, \pi]$. More expressive encodings (with entanglement, repeated data uploads, or hardware-efficient ansätze) can enrich the induced kernel beyond what is classically tractable.

A second, fundamentally different quantum strategy is the **quantum kernel method**: rather than optimising circuit parameters, the quantum circuit computes an inner product (a *kernel*) between pairs of encoded states,

$$\kappa(\mathbf{x}_i, \mathbf{x}_j) = \left| \langle \psi(\mathbf{x}_i) | \psi(\mathbf{x}_j) \rangle \right|^2, \quad (4)$$

and a classical SVM exploits this kernel to find the optimal hyperplane. In this formulation, *all learning happens classically*; the quantum device is used purely as a kernel evaluator. This makes quantum kernel methods immediately compatible with near-term hardware.

Importantly, the **small dataset regime** ($N \approx 30$) may be precisely where quantum methods shine: generalization bounds for variational quantum models scale as $\mathcal{O}(\sqrt{T \log T/N})$ [8], where T is the number of independently trainable gates. With few parameters and structured circuits, quantum models can be more data-efficient than classical counterparts.

5 Quantum Methods to Explore

The following quantum approaches are natural starting points. You are not required to implement all of them; depth is preferred over breadth.

1. Quantum Kernel SVM (QK-SVM) [3, 4]

Encode each drug’s features using a parameterized feature map $\mathcal{E}(\mathbf{x}; \boldsymbol{\theta})$, estimate the kernel matrix $K_{ij} = \kappa(\mathbf{x}_i, \mathbf{x}_j)$ (Eq. 4) via repeated circuit execution, and train a classical SVM on K .

- How does the choice of feature map (angle encoding, ZZ-feature map, hardware-efficient) affect accuracy?
- Does *Kernel Target Alignment* (KTA) help by optimising the feature map parameters to align K with the label structure?

2. Variational Quantum Classifier (VQC) [5]

Combine angle encoding with a trainable ansatz of parameterized rotation and entanglement layers. Minimise binary cross-entropy loss using a gradient-based optimizer.

- How many layers are sufficient before expressibility saturates on 29 samples?
- What is the effect of circuit depth on training stability (barren plateaus)?

3. Data Re-uploading Classifier [6]

Upload the classical features repeatedly at every layer of the circuit, interleaved with parameterized rotations. This gives the circuit universal approximation power without increasing qubit count.

- How many re-uploading layers are needed to converge?
- Compare the parameter-shift gradient to SPSA for this small dataset.

4. Quantum Convolutional Neural Network (QCNN) [7]

Apply hierarchical convolutional and pooling layers to a 6-qubit angle-encoded state, progressively reducing the qubit count until a single measurement qubit remains.

- A QCNN for 6 features reduces $6 \rightarrow 3 \rightarrow 2$ qubits via two pooling stages.
- A $SU(4)$ dense layer on the two final qubits captures all pairwise correlations.
- The QCNN has ~ 51 trainable parameters, well below the sample count $N = 29$.

Hint: Connecting quantum kernel and variational approaches

The quantum kernel method and the VQC are not entirely distinct. When the ansatz $U(\boldsymbol{\theta})$ acts on an encoded state $|\psi(\mathbf{x})\rangle$, the measurement outcome implicitly defines a kernel $\langle\psi(\mathbf{x}_i)|U^\dagger(\boldsymbol{\theta})OU(\boldsymbol{\theta})|\psi(\mathbf{x}_j)\rangle$. Optimising $\boldsymbol{\theta}$ in the VQC is thus equivalent to *learning the best kernel* for this dataset.

6 Challenge Tasks

Task 1: Dataset Exploration and Feature Engineering

Load the 29-drug dataset and explore the provided RDKit descriptor table before building any model.

- Visualise pairwise correlations between molecular features and COMDR₁₅.
- Identify the RDKit descriptors that best approximate the four variables used by Pätzmann et al. (D_{50} , $\log D_{6.5}$, κ_3 , S_{app}) and discuss why each one is physically relevant to dissolution. Note that D_{50} and S_{app} are experimental values provided directly; $\log D_{6.5}$ is not available from RDKit but $\log P$ (MolLogP) is a reasonable approximation; κ_3 is available directly as a topological descriptor.
- Select the features you will use for all subsequent tasks and justify your choice. Consider whether a broader RDKit descriptor set helps or hurts with only 29 samples.

Deliverable: Feature correlation heatmap; justified feature selection for subsequent tasks.

Task 2: Classical ML Baselines (minimum 2 models)

Establish your own classical reference using **at least two** different ML algorithms evaluated under LOOCV. You are free to choose your models; examples include Logistic Regression, SVM, Random Forest, k -NN, Gradient Boosting, and PLS regression.

- You may treat the problem as **regression** (predict continuous COMDR₁₅, report R^2 and Q^2) to enable a direct comparison with the Pätzmann benchmark, or as **classification** (predict Responder/Non-Responder, report accuracy, F1, and per-class recall), or **both**.
- Apply per-fold feature scaling inside the LOOCV loop to prevent data leakage.
- Summarise results in a single table covering all models.

Deliverable: LOOCV results table for all classical models.

Task 3: Quantum Feature Map Design

Before training any quantum model, analyse the geometry induced by your chosen encoding.

- Implement an angle-encoding circuit on 4 or 6 qubits (one R_y per feature).

- Compute and plot the 29×29 quantum kernel matrix as a heatmap, with rows and columns grouped by Responder/Non-Responder label.
- Compute the *Kernel Target Alignment* (KTA) score and discuss whether your feature map separates the two classes.
- Explore: does adding entanglement layers improve KTA?

Deliverable: Circuit diagram; kernel heatmap; KTA score.

Task 4: Quantum ML Models (minimum 2 models)

Implement at least **two** quantum ML approaches from Section 5 and evaluate both under full 29-fold LOOCV.

- Use Qiskit for all circuit construction and simulation.
- Apply per-fold `MinMaxScaler` to normalise features to $[0, \pi]$ inside the LOOCV loop; never fit the scaler on the held-out drug.
- For variational models, save the raw circuit output $P(|1\rangle)$ alongside the binary prediction: the continuous score preserves implicit regression information and allows ranking of drugs by predicted enhancement strength.
- Report the same metrics as your classical baseline table so all models are directly comparable.

Deliverable: LOOCV results table; per-drug score vs. true COMDR₁₅ scatter plot.

Task 5: Unified Comparison and Interpretation

Consolidate all results (classical and quantum) into a single comparison table and provide a critical analysis.

- Do your quantum models match, exceed, or fall short of your classical baselines?
- Compare your best result to the Pätzmann benchmark. Note that R^2 (regression) and classification accuracy are not directly comparable metrics; discuss how you translate between the two.
- Which drugs are hardest to predict correctly across *all* models? What physicochemical characteristics do these drugs share?
- Did the choice of feature set (4 Pätzmann features vs. a broader RDKit set) matter differently for classical vs. quantum models?

Deliverable: Unified results table; ≤ 1 -page discussion with chemical interpretation.

Bonus: Advanced Extensions

If time permits, explore one or more of the following:

- **Kernel Target Alignment optimisation:** tune the feature map parameters to maximise KTA prior to SVM training (trained quantum kernel).

- **Noise simulation:** run your best quantum circuit on a noise-aware backend (e.g., `AerSimulator` with a device noise model) and quantify the accuracy degradation.
- **Blind prediction:** a hidden set of drugs not included in the 29 will be used to evaluate true generalization. Submit a prediction (Responder / Non-Responder) and a confidence score for each hidden drug.

7 SDG Impact

This challenge aligns with three United Nations Sustainable Development Goals:

- **SDG 3: Good Health and Well-being.** Improving the predictability of drug formulation outcomes reduces the time and cost of bringing new medicines to patients. A reliable computational predictor of co-milling effectiveness could accelerate the development of better oral formulations for BCS Class II and Class IV drugs, which constitute the majority of the modern drug pipeline.
- **SDG 9: Industry, Innovation, and Infrastructure.** This challenge demonstrates how quantum computing can be applied to a real industrial formulation problem. It contributes to building a quantum-aware computational skill base in the pharmaceutical sciences and establishes a reproducible QML benchmark for dissolution prediction.
- **SDG 12: Responsible Consumption and Production.** Poor oral bioavailability is a major cause of drug attrition in late-stage development, contributing to enormous waste in resources, animal studies, and clinical trials. Accurate early-stage predictive models reduce this waste by enabling data-driven formulation decisions before expensive experiments are conducted.

8 Provided Resources

Participants will be provided with:

- **Dataset:** The experimental co-milling dissolution dataset (29 drugs, with COMDR_{15} , COMDR_{60} , particle size D_{50} , and measured apparent solubility in FaSSIF).
- **Molecular Descriptors:** A pre-computed table of RDKit descriptors for all 29 drugs, derived from their SMILES strings. The table covers more than 200 descriptors including Lipinski properties, topological indices, shape descriptors, and electronic properties. *The selection of which descriptors to use is part of the challenge:* not all descriptors are equally informative, and the choice must be justified.
- **Reference Literature:** The Pätzmann et al. paper [1] will be provided. Key QML references [3, 4, 6, 7] will be linked.

Software Stack

- **Python ≥ 3.10** with `numpy`, `pandas`, `scikit-learn`, `matplotlib`

- **Qiskit** ≥ 2.4 and **qiskit-machine-learning** ≥ 0.9
- **RDKit** ≥ 2026.3 (molecular descriptor computation)
- **qiskit-aer** (optional, for noisy simulation in the bonus task)

9 Target Audience and Prior Knowledge

This challenge is designed for participants at the **Intermediate** level, meaning familiarity with:

- Quantum circuits: single- and two-qubit gates, parameterized circuits
- Basic ML: classification, cross-validation, kernel methods
- Python and Qiskit (or willingness to learn)

No prior knowledge of pharmaceutical science or drug formulation is required. The dataset and problem are fully self-contained.

The challenge is particularly well-suited for:

- Quantum ML researchers interested in a *real, published, small-data* benchmark
- Computational chemistry students who want to apply quantum algorithms to molecular data
- QML practitioners who want to compare VQC, QK-SVM, and QCNN on the same problem

10 Evaluation Criteria

Submissions will be evaluated on the following dimensions:

- **Technical Correctness (30%)**: Is the LOOCV procedure implemented without data leakage? Are quantum circuits correctly defined and measured? Are metrics (accuracy, F1, per-class recall) reported correctly?
- **Quantum Understanding (25%)**: Does the team demonstrate understanding of the chosen quantum method? Can they articulate why a particular encoding or ansatz was selected? Is the kernel matrix or parameterized circuit meaningfully analysed?
- **Predictive Performance (20%)**: How does the best quantum model compare to the team’s own classical baselines? Does it approach or exceed the Pätzmann benchmark ($Q^2 = 0.77$)? Consistent improvement over the classical baselines with a sound justification constitutes an outstanding result; matching the classical baselines is a solid result.
- **Interpretability and Insight (15%)**: Does the analysis explain *why* the model succeeds or fails on specific drugs? Is the connection to chemical structure or the Pätzmann R^2 reference discussed?

- **Creativity and Depth (10%):** Bonus credit for attempting KTA optimisation, noise analysis, or the hidden test set. Innovative circuit designs, hybrid classical-quantum ensembles, or novel encoding strategies are particularly welcome.

Outstanding submissions will go beyond the starter code, demonstrate genuine understanding of the quantum-pharmaceutical link, and produce insights that inform the broader question: *is there a regime where QML outperforms classical ML on real molecular data?*

References

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